

Statistical Methods for Experimental Physics (Part-I)

Problem Set

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1 Frequentist Statistics Problems

These problems are entirely optional and will not be assessed. You might find them useful to prepare for the Exam or to check your understanding of the material.

1.1 Kolmogorov axioms

Show that the definition of *frequentist* probability given in the lectures satisfies the three Kolmogorov axioms (I don't expect any very rigorous proofs). For this question, you can always assume we are concerned with the probabilities related to *exclusive outcomes*.

Extra Challenge If you can also show this for the Bayesian definition, then have a go!

ANSWER:

Frequentist probability. The definition of frequentist probability given in the lectures is that if counting a particular outcome X as n from a total number of trials N , the ratio n/N converges to the probability for that outcome to occur - or,

$$\lim_{N \rightarrow \infty} \frac{n}{N} = P(X)$$

- For the first axiom, clearly both n and N are positive, so the ratio n/N must also be positive for any n . It follows that $P(X) \geq 0$ for any X .
- The second axiom is also satisfied since if one counts either X or Y occurrences in N trials as n_X and n_Y , the total number of outcomes either X or Y must be $n_X + n_Y$. If the limits exist, then from the definition of limits, we know,

$$P(X \text{ or } Y) = \lim_{N \rightarrow \infty} \frac{n_X + n_Y}{N} = \lim_{N \rightarrow \infty} \frac{n_X}{N} + \lim_{N \rightarrow \infty} \frac{n_Y}{N} = P(X) + P(Y)$$

- Finally, the third axiom must be satisfied since if one counts any possible outcome X_i as n_{X_i} from N trials, the probability of any outcome to occur must be

$$P(\text{any } X_i) = \lim_{N \rightarrow \infty} \frac{\sum_i n_{X_i}}{N}.$$

But since $\sum_i n_{X_i} = N$, $P(\text{any } X_i) = \lim_{N \rightarrow \infty} \frac{N}{N} = 1$.

Bayesian probability. For Bayesian probability, we can use the concept of betting odds. Its easiest to show that the Bayesian definition of probability satisfies the axioms by using a simple game, rolling a six-sided die. Remember we define the probability of a win by,

$$P(W) = \frac{B}{B + W},$$

where B is the amount we bet and W is the *minimum* amount of *profit* we would be happy to win for a bet of B - i.e $B + W$ is the amount we need to *win* such that in the long run (or across different games) we'd expect to break even.

- For the first axiom, clearly the amount we bet or win should be positive so the ratio $B/(B + W)$ must be positive for any B . It follows that $P(W) \geq 0$.
- For the second axiom, let's say we define two games based on rolling a six-sided die. In the first game, we win if the die rolls a 1. In this case, we would need to win £5 for a bet of £1 when we roll a 1. The probability is then $P(1) = \frac{1}{1+5} = \frac{1}{6}$. In a second game, if the die rolls greater or equal to 5, we win and in this case, we would expect $W = £2$ for $B = £1$ and $P(\geq 5) = \frac{1}{1+2} = \frac{1}{3}$. If we combine the games such that we win on either a roll

of 1 or a roll of greater than 5, we would expect to $W = £1$ for $B = £1$ to break even (have a think about why). Note that we cannot roll both 1 and greater than 5 in the same game so these are exclusive outcomes. If we sum the probabilities from the individual games, we get, $\frac{1}{6} + \frac{1}{3} = \frac{1}{2} = \frac{1}{1+1} = P(1 \text{ or } \geq 5)$.

- For the third axiom, we are asking how much we would expect to profit if we win on any outcome of the die roll. Clearly, we could ask for 0 profit and still break even in the long run. We have,

$$P(\text{anything}) = \frac{1}{1+0} = 1.$$

1.2 Zero correlation does not mean independence

In the lectures, we said that two random variables which are independent will have a zero correlation coefficient.

1. Show that two continuous random variables X and Y , with $(X, Y) \sim f(X, Y)$ which are independent will have a 0 correlation coefficient.
2. Let X be a continuous random variable symmetrically distributed around 0 with a density function $f(X)$. Let $Y = X^2$. Show that despite the fact that Y and X are clearly dependent, their correlation coefficient is 0.

ANSWER:

Part 1 The first thing to remember is that if X and Y are independent then their probability distributions can be decomposed into $f(X, Y) = g(X)h(Y)$. The expectation value of (X, Y) is,

$$E[XY] = \int \int XY \cdot f(X, Y) dX dY = \int \int XY \cdot h(X)g(Y) dX dY = \int Xh(X)dX \int Yg(Y)dY = E[X]E[Y].$$

From the lectures, we know that the definition of the correlation coefficient is,

$$\rho(X_i, X_j) = \frac{\nu_{1,1}^{ij}}{\sqrt{\nu_{2,2}^i \nu_{2,2}^j}} = \frac{E[(X_i - E[X_i])(X_j - E[X_j])]}{\sqrt{E[(X_i - E[X_i])^2]E[(X_j - E[X_j])^2]}}$$

Looking at the numerator of the right hand side of this expression, if $\nu_{1,1}^{ij} = 0$, then $\rho(X_i, X_j) = 0$. Multiplying out the outer expectation in the numerator, we have,

$$E[(X_i - E[X_i])(X_j - E[X_j])] = E[X_i X_j - X_i E[X_j] - X_j E[X_i] + E[X_i]E[X_j]],$$

and by the linearity property of the expectation $E[\cdot]$ and the fact that $E[X_i]$ and $E[X_j]$ are constant numbers, we have,

$$E[X_i X_j - X_i E[X_j] - X_j E[X_i] + E[X_i]E[X_j]] = E[X_i X_j] - E[X_i]E[X_j] - E[X_j]E[X_i] + E[X_i]E[X_j]E[1] = E[X_i X_j] - E[X_i]E[X_j].$$

We already showed that $E[XY] = E[X]E[Y]$ if X and Y are independent, so we have that the correlation coefficient is zero.

Part 2 Again we can calculate just the numerator ($\nu_{1,1}^{ij}$) of the correlation coefficient formula. We have,

$$\nu_{1,1}^{X, X^2} = E[X \cdot X^2] - E[X]E[X^2].$$

Since X is *symmetrically distributed around 0*, the expectation of X is 0, and so is any odd function (any function such that $g(-X) = -g(X)$) of X such as $f(X) = X^3$. We can see this since the expectation value of the function $g(X)$ is given by,

$$\begin{aligned} \int_{-\infty}^{\infty} g(X)f(X)dX &= \int_{-\infty}^0 g(X)f(X)dX + \int_0^{\infty} g(X)f(X)dX \\ &= \int_{-\infty}^0 g(X)f(-X)dX + \int_0^{\infty} g(X)f(X)dX \\ &= -\int_{-\infty}^0 g(-X)f(-X)dX + \int_0^{\infty} g(X)f(X)dX \\ &= \int_{-\infty}^0 g(-X)f(-X)(-dX) + \int_0^{\infty} g(X)f(X)dX \\ &= \int_{-\infty}^0 g(X)f(X)dX + \int_0^{\infty} g(X)f(X)dX \\ &= -\int_0^{\infty} g(X)f(X)dX + \int_0^{\infty} g(X)f(X)dX \\ &= 0 \end{aligned}$$

where we've used the fact that $f(X) = f(-X)$ i.e it is the (symmetric or even) probability distribution of X . Therefore, $\nu_{1,1}^{X, X^2}$ and hence the correlation between X and X^2 is zero. Clearly these two are very much dependent so this example shows how zero correlation does not necessarily imply independence between two variables.

1.3 Moments of the Poisson distribution

In the lecture notes, we calculated the expectation and variance of the Poisson distribution with parameter λ are $E[k] = \lambda$ and $V(k) = \lambda$.

What is the third central moment of the Poisson distribution? You should show how you derive your answer.

ANSWER: In the lecture notes, we had an example where we calculated the expectation and variance of the Poisson distribution. We used the following identities there;

$$\sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} = (\lambda^2 + \lambda)e^{\lambda}$$

and

$$\sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} = \lambda e^{\lambda}$$

The definition of the *third central moment* of a discrete random variable k with probability distribution $p(k)$ was defined in lectures to be,

$$E[(k - E[k])^3] = \sum_k (k - E[k])^3 p(k).$$

For the case where $p(k)$ is a Poisson distribution with parameter λ this becomes,

$$E[(k - E[k])^3] = \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!},$$

where we used the fact that $E[k] = \lambda$ for the Poisson case, as shown in the lectures.

Now, lets expand the brackets,

$$\begin{aligned} E[(k - E[k])^3] &= \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &= \sum_{k=0}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &\quad + 3\lambda^2 \sum_{k=0}^{\infty} k \lambda^k \frac{e^{-\lambda}}{k!} \\ &\quad - 3\lambda \sum_{k=0}^{\infty} k^2 \lambda^k \frac{e^{-\lambda}}{k!} \\ &\quad - \lambda^3 \sum_{k=0}^{\infty} \lambda^k \frac{e^{-\lambda}}{k!}. \end{aligned}$$

The last term is easy since this is just $\lambda^3 \sum_k p(k) = \lambda^3$ as the total probability must equal 1. For the 2nd and 3rd terms, we can use the identities above to simplify them so that we are left with,

$$\begin{aligned} E[(k - E[k])^3] &= \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &= \sum_{k=0}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &\quad + 3\lambda^2 \times \lambda \\ &\quad - 3\lambda \times (\lambda^2 + \lambda) \\ &\quad - \lambda^3. \end{aligned}$$

For the first term then, we need to think a bit more. We can notice however that,

$$\begin{aligned} \sum_{k=0}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} &= 0 + \sum_{k=1}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &= \sum_{k=1}^{\infty} (k \times k^2) (\lambda \times \lambda^{k-1}) \frac{e^{-\lambda}}{k \times (k-1)!} \\ &= \lambda \sum_{k=1}^{\infty} k^2 \lambda^{k-1} \frac{e^{-\lambda}}{(k-1)!} \\ &= \lambda \sum_{p=0}^{\infty} (p+1)^2 \lambda^p \frac{e^{-\lambda}}{p!}, \end{aligned}$$

where we've made the substitution $k \rightarrow p + 1$ in the last line. Now we can expand the brackets to get,

$$\lambda \sum_{p=0}^{\infty} (p+1)^2 \lambda^p \frac{e^{-\lambda}}{p!} = \lambda \left(\sum_{p=0}^{\infty} p^2 \lambda^p \frac{e^{-\lambda}}{p!} + \sum_{p=0}^{\infty} 2p \lambda^p \frac{e^{-\lambda}}{p!} + \sum_{p=0}^{\infty} \lambda^p \frac{e^{-\lambda}}{p!} \right),$$

and once again we can simplify using our identities,

$$\lambda \sum_{p=0}^{\infty} (p+1)^2 \lambda^p \frac{e^{-\lambda}}{p!} = \lambda (\lambda^2 + \lambda + 2\lambda + 1) = \lambda^3 + 3\lambda^2 + \lambda,$$

and finally we can substitute this back into our expression for the third moment to get to,

$$\begin{aligned} E[(k - E[k])^3] &= \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &= \lambda^3 + 3\lambda^2 + \lambda + 3\lambda^3 - 3\lambda^3 - 3\lambda^2 - \lambda^3 \\ &= \lambda, \end{aligned}$$

so the third central moment of a Poisson distribution is λ .

Alternative Answer: Another way of finding the answer is to use moment generating functions and cumulant generating functions. We didn't discuss these at all in the lectures but if you already know about them, it's fine to use this method for this question.

The moment generating function for a Poisson random variable is given by,

$$m(t) = e^{\lambda(e^t - 1)} \quad (1)$$

and the cumulant generating function is given by the logarithm of the moment generating function, so for a Poisson it is,

$$c(t) = \lambda(e^t - 1). \quad (2)$$

It is generally true that the first algebraic moment and, second and third central moments are equal to the first, second and third derivatives of the cumulative generating function evaluated at $t = 0$, so we have that,

$$\nu_3 = \frac{d^3 c(t)}{dt^3} \Big|_{t=0} = \lambda e^0 = \lambda. \quad (3)$$

1.4 Cauchy distribution

In the lectures, we showed how the sum of two Gaussian distributed random variables is itself Gaussian distributed. Suppose now that $X \sim \phi(X; 0, 1)$ and $Y \sim \phi(Y; 0, 1)$ are independent random variables. Show that the distribution of $U = \frac{X}{Y}$ is Cauchy, i.e that $p(U) = \frac{1}{\pi(1 + U^2)}$.

Hint: You should start by deriving the marginal distribution formula for the ratio of two independent random variables. Be careful that the case $Y = 0$ will cause a problem, so split the marginal distribution into two cases, one for $Y > 0$ and one for $Y < 0$. The sum of these marginal distributions will be the total distribution.

ANSWER: As the hint in the question suggested, we start by deriving the marginal distribution for the ratio of our two independent variables. Let $U = \frac{X}{Y}$ and $V = Y$ as our change of variables. For $Y \neq 0$, our infinitesimal volume transforms as,

$$dXdY = \left| \det \begin{bmatrix} \frac{\partial U}{\partial X} & \frac{\partial U}{\partial V} \\ \frac{\partial V}{\partial X} & \frac{\partial V}{\partial V} \end{bmatrix} \right|^{-1} dUdV = \left| \det \begin{bmatrix} \frac{1}{Y} & -\frac{X}{Y^2} \\ 0 & 1 \end{bmatrix} \right|^{-1} dUdV = Y \cdot dUdV = V \cdot dUdV.$$

We calculate the marginal distribution $p(U)$ first for values of V greater than zero. We have,

$$\begin{aligned} p(U)^+ &= \int_{\epsilon}^{\infty} h(U, V) dV = \int_{\epsilon}^{\infty} \phi(UV) \phi(V) V dV \\ &= \int_{\epsilon}^{\infty} \frac{1}{2\pi} e^{-\frac{U^2 V^2}{2}} e^{-\frac{V^2}{2}} V dV \\ &= \int_{\epsilon}^{\infty} \frac{1}{2\pi} e^{-\frac{V^2(1+U^2)}{2}} V dV \\ &= -\frac{1}{2\pi} \left[\frac{1}{1+U^2} e^{-\frac{V^2(1+U^2)}{2}} \right]_{\epsilon}^{\infty} \\ &= \frac{1}{2\pi(1+U^2)} e^{-\frac{\epsilon^2(1+U^2)}{2}} \end{aligned}$$

where $\epsilon > 0$ is some negligible number and the superscript $+$ means we only take the part of the probability distribution from the region where $Y = V > 0$. In the limit of $\epsilon \rightarrow 0$, we get,

$$p(U)^+ = \frac{1}{2\pi(1+U^2)}.$$

Similarly, we can take the case where $Y = V < 0$ (for $p(U)^-$ and we will get an identical result so that,

$$p(U) = p(U)^- + p(U)^+ = \frac{1}{\pi(1+U^2)},$$

which is the Cauchy distribution.

1.5 The Lying Friend

A friend of yours is known to only tell the truth with a probability $0 \leq x \leq 1$.

- i Suppose you have a deck of 52 cards and this friend draws one of the cards and declares that it's a King. What is the probability that the card is a King? Express the answer in terms of x .
- ii Plot this probability as a function of x . What would x need to be before you are more than 50% sure that the card is a King?

Hint: A standard deck of cards contains 4 Kings. You can use Bayes' rule to solve this question.

ANSWER: For part i: Let the following probabilities be defined;

- $P(E)$ = The probability that your friend says that the card is a King.
- $P(K)$ = The probability that a random card drawn is a King.
- $P(E|K)$ = The probability that your friend says the card is a King when the card really is a King.
- $P(K|E)$ = The probability that the card is a King given that your friend has said that its a King.

Using Bayes' rule we have that

$$P(K|E) = \frac{P(E|K)P(K)}{P(E)}$$

Remember that $P(E)$ is the probability says the card is a King regardless of what the truth is. This is the sum of the probability that your friend says the card is a King when it is a King multiplied by the probability that it is a King ($P(E|K) \times P(K)$), and the probability that your friend says the card is a King when it is not a King ($P(E|\neg K)P(\neg K) = (1-x) \times (1-P(K))$), so,

$$P(E) = P(E|K) \times P(K) + P(E|\neg K)P(\neg K) = xP(K) + (1-x)(1-P(K)),$$

and we have,

$$P(K|E) = \frac{xP(K)}{xP(K) + (1-x)(1-P(K))} = \frac{x \frac{4}{52}}{x \frac{4}{52} + (1-x) \frac{48}{52}},$$

where we've subbed in $P(K) = \frac{4}{52}$.

For part ii: You should get something like the Figure below, Setting $P(K|E) = 0.5$ and re-arranging our answer to part i, we find,

$$x = \frac{48}{52}$$

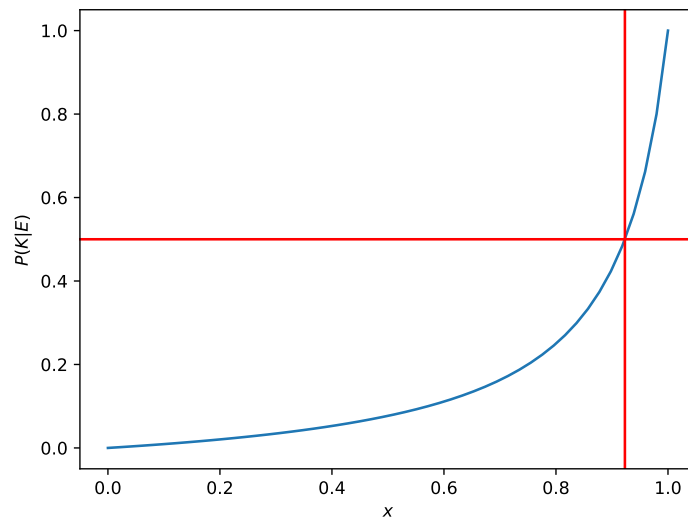


Figure 1: Probability that the card drawn is a King given that your (sometimes lying) friend says it's a King. The lines show the value of x at which the probability that the card is a King is at least 50%.

1.6 Convergence to a uniform distribution

Let X_n be a random variable with probability density function $f_n(X)$ given by,

$$f_n(X) = 1 - \cos 2n\pi X$$

for $X \in [0, 1]$ and $f_n(X) = 0$ otherwise. Show that X_n converges in distribution to a random variable X which is uniformly distributed on $[0, 1]$

ANSWER:

To show that X_n converges in distribution to a uniform distribution on $[0, 1]$, we need to show that the cumulative distribution function (CDF) of X_n , $F_n(X)$ converges to the CDF of a uniform distribution on $[0, 1]$, which is given by,

$$F(X) = \begin{cases} 0 & X < 0 \\ X & 0 \leq X \leq 1 \\ 1 & X > 1 \end{cases}$$

The CDF of X_n is given by,

$$F_n(X) = \int_0^X f_n(t) dt = \int_0^X (1 - \cos 2n\pi t) dt = \begin{cases} 0 & X < 0 \\ X - \frac{\sin 2n\pi X}{2n\pi} & 0 \leq X \leq 1 \\ 1 & X > 1 \end{cases}$$

Now as $n \rightarrow \infty$, the term $\frac{\sin 2n\pi X}{2n\pi}$ oscillates between -1 and 1, but its amplitude decreases to 0 as there is a $1/n$ term. Therefore, for any fixed X in the interval $[0, 1]$, we have that

$$\lim_{n \rightarrow \infty} F_n(X) = X - 0 = X$$

for $0 \leq X \leq 1$. For the other two cases, it is clear that $F_n(X)$ converges to $F(X)$ as well since they are constant functions. The figure below shows the probability density and cumulative distribution functions for different values of n .

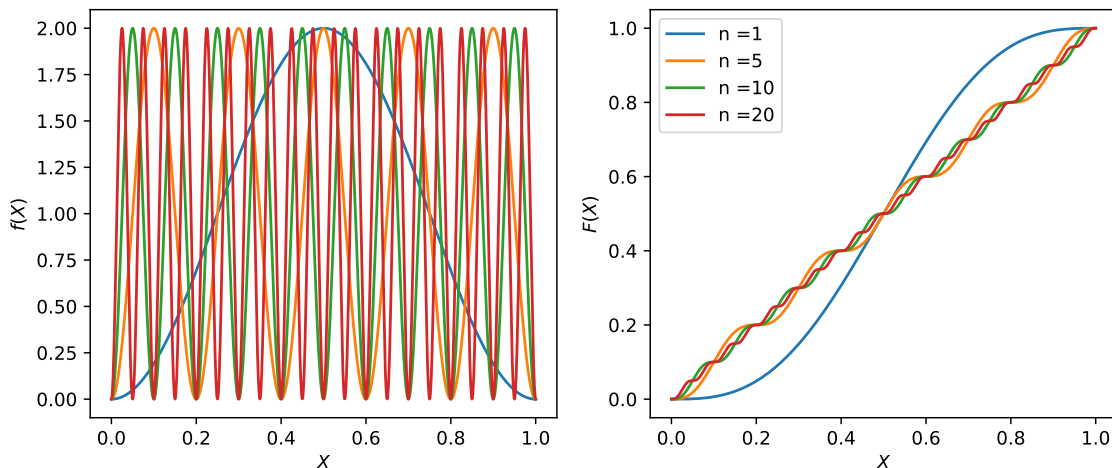


Figure 2: Probability density function for f_n (left) and its cumulative distribution function (right) for increasing values of n .

1.7 Convergence to a Dirac-delta function

Let $X_n \sim \phi(X; 0, 1/n)$ for each n . Show that X_n converges in distribution to a probability density function which is a Dirac delta function at 0 – $\delta(0)$.

Hint: To show this, show that for any $x < 0$, $F_n(x) \rightarrow 0$, and for any $x > 0$, $F_n(x) \rightarrow 1$, as $n \rightarrow \infty$. Providing an answer using a numerical method is sufficient to get the full marks (feel free to write some code to demonstrate this, and attach plots to prove it to me!).

ANSWER: For this question, starting with the hint, we calculate the cumulative distribution function $F_n(X)$,

$$F_n(X) = \int_{-\infty}^X \frac{n}{\sqrt{2\pi}} e^{-\frac{t^2 n^2}{2}} dt.$$

This integral is not possible to express in closed form, however we can use the fact that the anti-derivative of a Gaussian function is related to the erf function via,

$$\int \frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2} \text{erf} \left(\frac{x-\mu}{\sigma\sqrt{2}} \right),$$

to get,

$$F_n(X) = \begin{cases} \frac{1}{2} + \left[\frac{1}{2} \operatorname{erf} \left(\frac{Xn}{\sqrt{2}} \right) \right]_0^X & X > 0 \\ 1 - \frac{1}{2} + \left[\frac{1}{2} \operatorname{erf} \left(\frac{Xn}{\sqrt{2}} \right) \right]_0^X & X < 0, \end{cases}$$

where we've used the fact that $\int_{-\infty}^{\infty} F_n(X) = 1$ and that $\phi_n(X; 0, 1/n)$ is symmetric about 0. In fact, both cases are identical (have think why, based on the integrals that we're performing). We can use the `scipy.special.erf` function to plot how this evolves as $n \rightarrow \infty$. This is shown in Figure ?? . It's easy to see that as $n \rightarrow \infty$, we have,

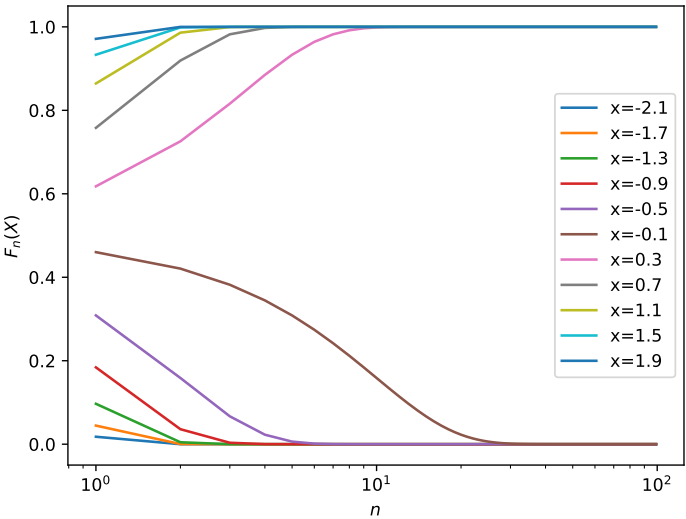


Figure 3: $F_n(X)$ for different values of X and increasing values of n .

$$F_n(X) \rightarrow F(X) = \begin{cases} 1 & X > 0 \\ 0 & X < 0, \end{cases}$$

This is also obvious just plotting the pdf and cdf of the Gaussian functions for increasing n , as shown in Figure ?? . I will give the points for any method that demonstrates the convergence, including numerical demonstrations.

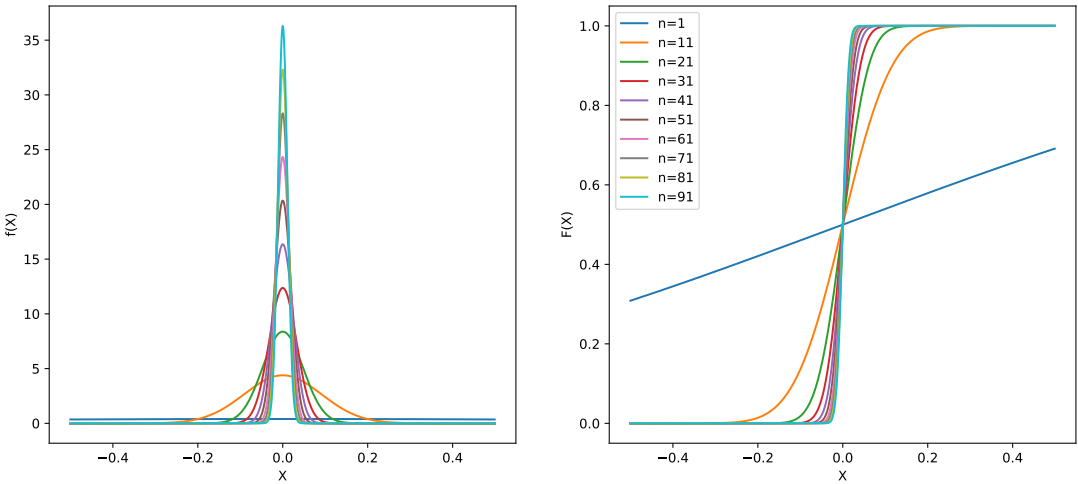


Figure 4: Probability density function for ϕ_n (left) and its cumulative distribution function (right) for increasing values of n .

As mentioned in the hint, providing an answer using a numerical method (as I did above) is sufficient to get the full marks.

1.8 The Maxwell-Boltzmann distribution

The Maxwell-Boltzmann distribution describes the distribution of the speed, $v \in [0, +\infty)$, of molecules in a gas at temperature T . Its probability density is given by,

$$f(v; T) = \left(\frac{m}{2\pi kT} \right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2kT}},$$

where m is the molecular mass and k is a constant known as the Boltzmann constant. The speed v of any molecule will be a random variable $v \sim f(v; T)$.

- i. Show that the first algebraic moment (the expectation of v) of the Maxwell-Boltzmann distribution μ_1 is given by,

$$\mu_1 = E[v] = 2\sqrt{\frac{2kT}{m\pi}}.$$

You may use the following identity,

$$\int_0^{+\infty} x^3 e^{-bx^2} dx = \frac{1}{2b^2}.$$

ANSWER: The first algebraic moment is defined by,

$$E[v] = \int_0^{+\infty} f(v; T) v dv = \int_0^{+\infty} \left(\frac{m}{2\pi kT}\right)^{3/2} 4\pi v^3 e^{-\frac{mv^2}{2kT}} dv.$$

Noting that the integral is now in the form given in the identity, we set $b = m/2kT$ and using the identity we have that,

$$E[v] = \left(\frac{b}{\pi}\right)^{3/2} 4\pi \int_0^{+\infty} v^3 e^{-bv^2} dv = \left(\frac{b}{\pi}\right)^{3/2} \frac{4\pi}{2b^2} = 2\sqrt{\frac{1}{\pi b}} = 2\sqrt{\frac{2kT}{m\pi}}.$$

- ii. Imagine an experimental apparatus can measure speeds of molecules in a gas v and observes a number n of speeds $\{v_1, v_2, \dots, v_n\}$. The observations are independent and identically distributed random variables distributed as $v_i \sim f(v; T)$.

Show that the sample mean, $\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$, is an *unbiased estimator* for the first algebraic moment of the the Maxwell-Boltzmann distribution.

ANSWER: The first algebraic moment is just the expectation value from the previous question so we set that equal to the sample mean to estimate it,

$$\hat{\mu}_1 = \bar{v} = \frac{1}{n} \sum_{i=1}^n v_i.$$

The definition of bias is the difference between the expectation value of the estimator and the true value.

$$\text{Bias}(\hat{\mu}_1) = E[\hat{\mu}_1] - \mu_1 = E\left[\frac{1}{n} \sum_{i=1}^n v_i\right] - \mu_1.$$

This can either be written as above or in words “if the estimator is unbiased, then the expectation value of the estimator is equal to the true value”. By the linearity of the expectation operator, we have

$$E\left[\frac{1}{n} \sum_{i=1}^n v_i\right] = \frac{1}{n} \sum_{i=1}^n E[v_i],$$

and since $E[v_i] = \mu_1$, we have that,

$$E\left[\frac{1}{n} \sum_{i=1}^n v_i\right] = \frac{1}{n} \sum_{i=1}^n \mu_1 = \mu_1 \frac{n}{n} = \mu_1,$$

since μ_1 is a constant. Plugging this into the equation for the bias above, we find that $\text{Bias}(\hat{\mu}_1) = \mu_1 - \mu_1 = 0$ - i.e the estimator is unbiased.

1.9 Phone call for Peter Higgs

You are sat at home with your statistics problems staring at you from across the room, and rather than solving them, you decide to procrastinate by dialing random telephone numbers. You punch a phone number in only to have Peter Higgs himself pick up at the other end. What is the probability of picking Professor Higgs’ number out of all possible phone numbers with an Edinburgh area code? Is this more or less likely than the discovery of the Higgs boson being an accident due to a large fluctuation of the background?

Hint: You can assume the Higgs discovery was at exactly 5σ (or sometimes said “5 standard deviations”). Convert this to a p -value using a one-sided test as explained in lectures. Also assume the Edinburgh phone number can be any 11 digit number starting with 0131.

ANSWER: For this problem, note that we assume all of the phone numbers between 01310000000 and 01319999999 are valid numbers (and that Peter Higgs has an Edinburgh landline phone of course!). This means there are $10,000,000 = 10^7$ possible phone numbers meaning the probability that you randomly dial Peter Higgs’ number is 10^{-7} . For the p -value calculation, we can use,

$$p = 1 - \int_{-\infty}^5 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \approx 2.867 \times 10^{-7},$$

so randomly calling Peter Higgs’ number is roughly the same (only 3 times less) probability as observing a p -value at least as small as the one we observed at the time of discovery. For the integral, I used the function `scipy.special.erf` which is related to the above p -value.

```

1 from scipy.special import erf
2 Z = 5.
3 p = 1-0.5*(1+erf(Z/(2**0.5)))

```

These days, in some channels we have well above 7σ in terms of the signal significance for the Higgs boson, which corresponds to a p -value smaller than 1.28×10^{-12} !

1.10 Maximum likelihood for exponential decay

Suppose we want to determine the lifetime (τ) of an unstable particle. The probability for this particle to decay at time t is given by,

$$p(t) = N(\tau)e^{-t/\tau} \quad (4)$$

Suppose then that in the lab, we've made measurements of the decay times (say by measuring the times at which we detect one or more decay products from a source), and have the following observed decay times; $t = 7.44, 4.13, 26.85, 1.42, 3.46, 2.68, 4.1, 4.04, 7.9$ and 12.03 in ns.

- i. Normalise the probability density function (i.e what is $N(\tau)$?)

ANSWER: To normalise the probability distribution, remember that,

$$\int_0^\infty p(t)dt = 1.$$

We have then that,

$$1 = \int_0^\infty N(\tau)e^{-t/\tau}dt = N(\tau) \left[-\tau e^{-t/\tau} \right]_0^\infty = N(\tau) (-0 + \tau) = N(\tau)\tau.$$

so $N(\tau) = \frac{1}{\tau}$.

- ii. What does the log-likelihood function look like for this dataset? Use your favourite plotting program to plot it as a function of τ .

ANSWER: The log likelihood function is given by,

$$\ln L(\tau) = \ln \prod_i p(t_i; \tau) = \sum_i \ln p(t_i; \tau) = \sum_i \ln \left(\frac{1}{\tau} e^{-t_i/\tau} \right) = -10 \ln(\tau) - \sum_i \frac{t_i}{\tau},$$

where i runs over the dataset of decay times $\{t_i\}_{i=1}^{10}$. A plot of $\ln L(\tau)$ is shown in Figure ??.

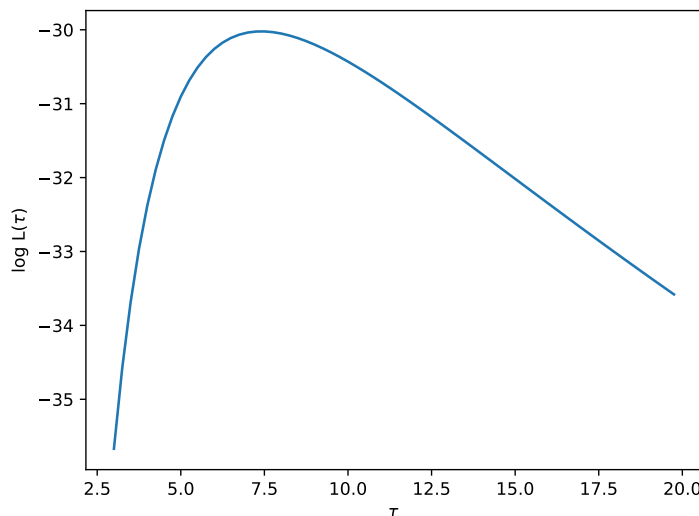


Figure 5: Plot of $\ln L(\tau)$ for τ .

- iii. What is the maximum likelihood (or minimum negative log-likelihood) estimator for τ ? **Hint:** You can calculate this analytically or numerically (or even better, do both and compare answers!).

ANSWER: The minimum negative log-likelihood estimator for τ can be found by solving,

$$0 = \frac{d(-\ln L(\tau))}{d\tau} \Big|_{\tau=\hat{\tau}} = 10 \frac{d}{d\tau} (\ln(\tau)) \Big|_{\tau=\hat{\tau}} + \sum_i \frac{d}{d\tau} \left(\frac{t_i}{\tau} \right) \Big|_{\tau=\hat{\tau}} = \frac{10}{\hat{\tau}} - \sum_i \frac{t_i}{\hat{\tau}^2},$$

and multiplying through by $\hat{\tau}$ and re-arranging we get,

$$\hat{\tau} = \frac{1}{10} \sum_i t_i.$$

The right hand side of this equation is the sample mean of the decay times so the estimator for τ is just the sample mean. In this case, for our data, the result is $\hat{\tau} = 7.405$ ns. We can also calculate this numerically (see the code below) and we'll get a very similar answer.